

Basic Code Analysis: Dockerfile Security & Best Practice Audit

Duration : 30 Min.

Scenario

A container security scanner flagged a production Dockerfile for running as the root user. Audit and create a secure Dockerfile in student_workspace/.

Target File Location & Creation

File to Create/Update: student_workspace/Dockerfile

Task Objectives

Perform the following actions inside the student_workspace directory: - Create a file named 'Dockerfile' directly inside 'student_workspace/'. Add a non-root user instruction ('USER node' or 'USER appuser') before CMD/ENTRYPOINT.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect the scenario input files in student_workspace/.
3. Create/update the file student_workspace/Dockerfile and populate it with valid parameters and configurations.
4. Save your changes (Ctrl + S or Cmd + S).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your work by running python run.py in the terminal before submitting.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Dockerfile exists in student_workspace/	3 Marks

TC2	Non-root user creation (USER node / USER appuser) added	4 Marks
TC3	Root privileges dropped before CMD/ENTRYPOINT	3 Marks

Total Score: 10 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.